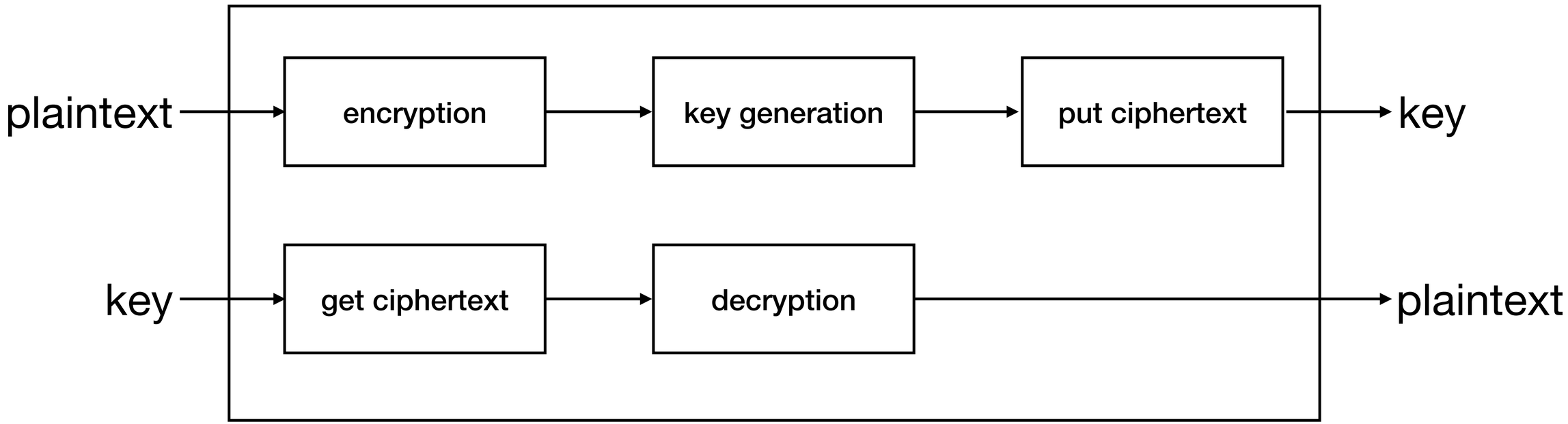
최종 프로젝트

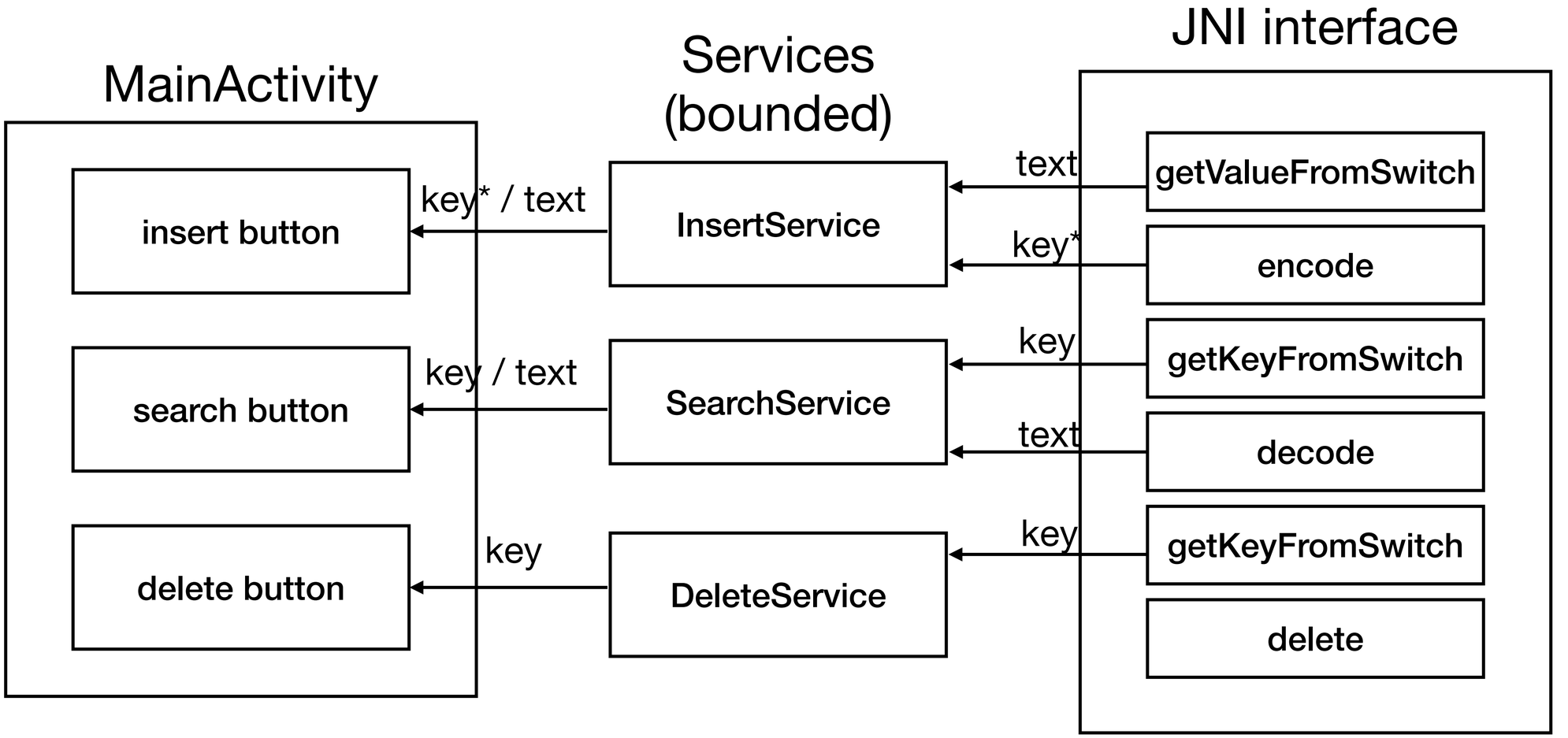
텍스트 암호화 시스템

이승우

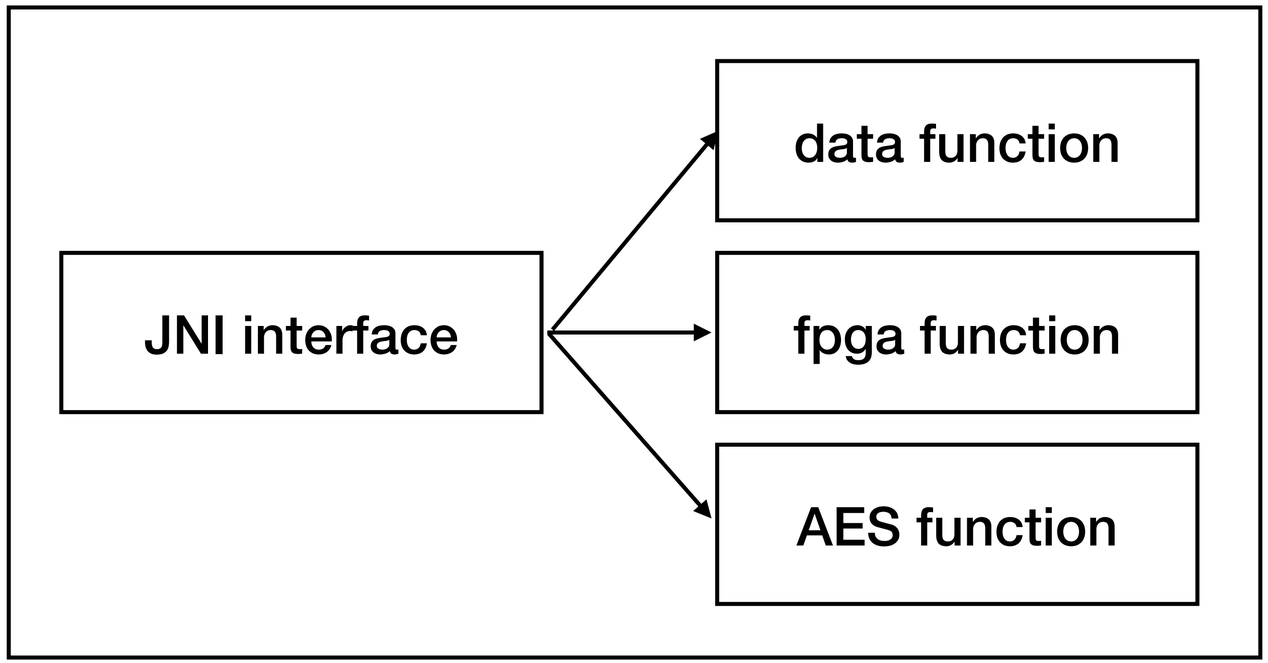
시스템 소개


plaintext
encryption
key generation
put ciphertext
key
key
get ciphertext
decryption
plaintext

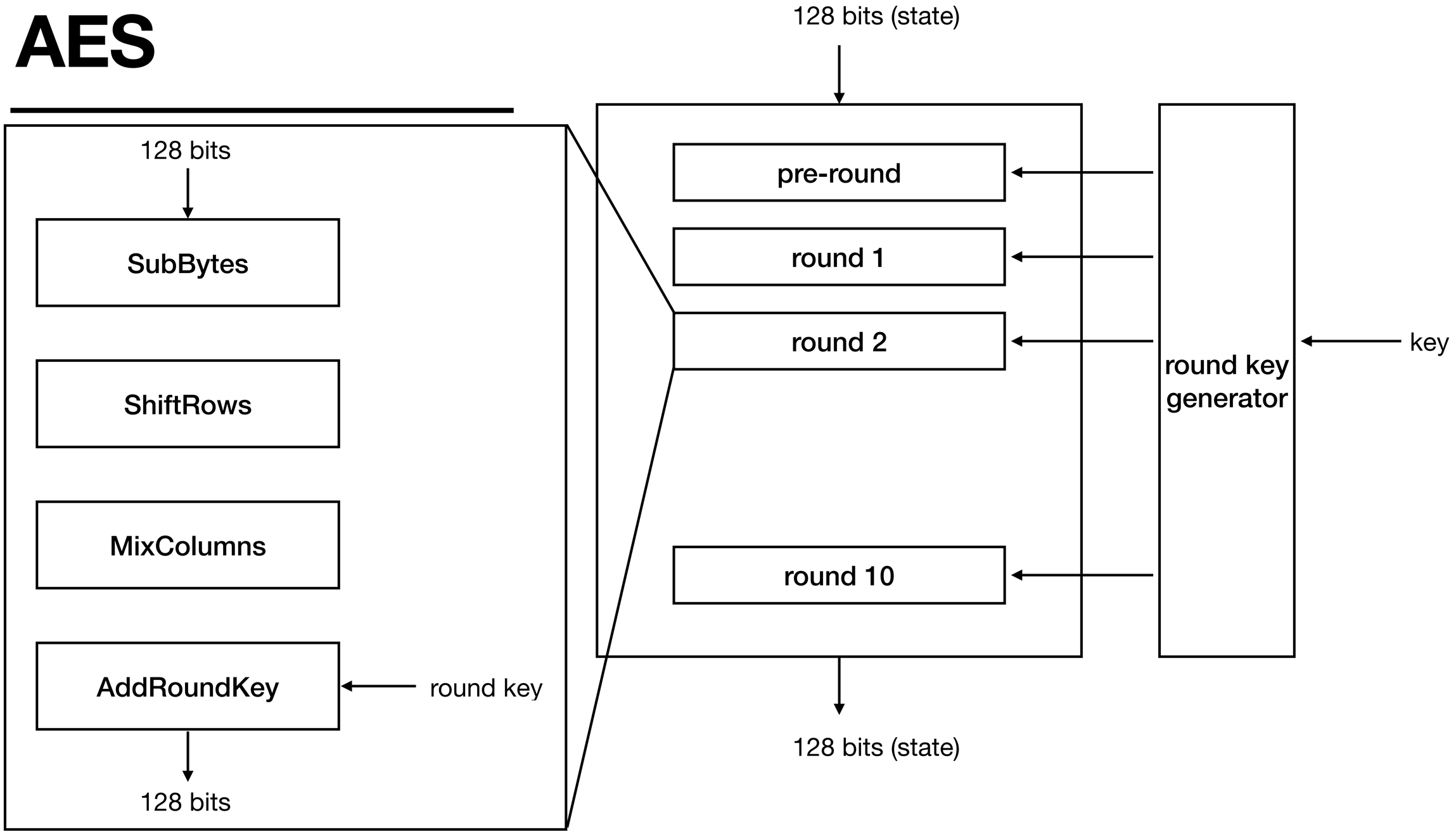
안드로이드 구조


MainActivity
Services
(bounded)
JNI interface
insert button
search button
delete button
key* / text
key / text
key
InsertService
SearchService
DeleteService
text
key*
key
text
key
getValueFromSwitch
encode
getKeyFromSwitch
decode
getKeyFromSwitch
delete

JNI



AES


128 bits (state)
pre-round
round 1
round 2
round 10
round key generator
key
128 bits (state)
128 bits
SubBytes
ShiftRows
MixColumns
AddRoundKey
round key
128 bits

JNI

AES

```
#include <linux/unistd.h>
#include <sys/syscall.h>

#define ENCRYPTION(plain, cipher) syscall(379, plain, cipher, 0)
#define DECRYPTION(plain, cipher) syscall(379, plain, cipher, 1)
```

JNI

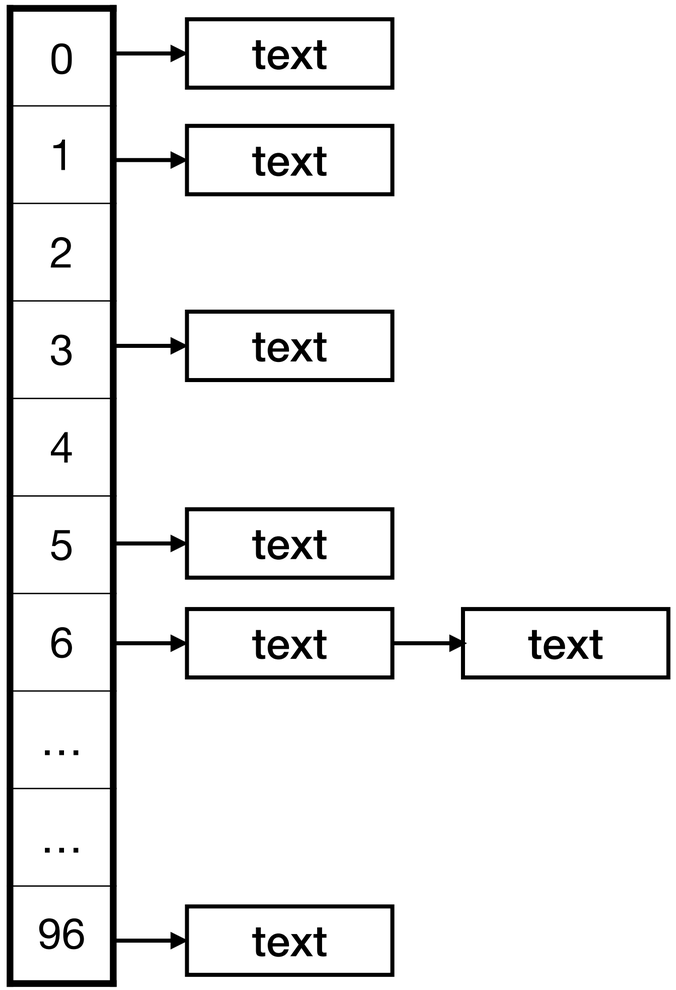
generateKey

```
unsigned int
generateKey (char *value) {
  unsigned int key = 0;

  while (*value) {
    key = (key << 3) + *value++;
  }
  return key % KEY_SIZE;
}
```

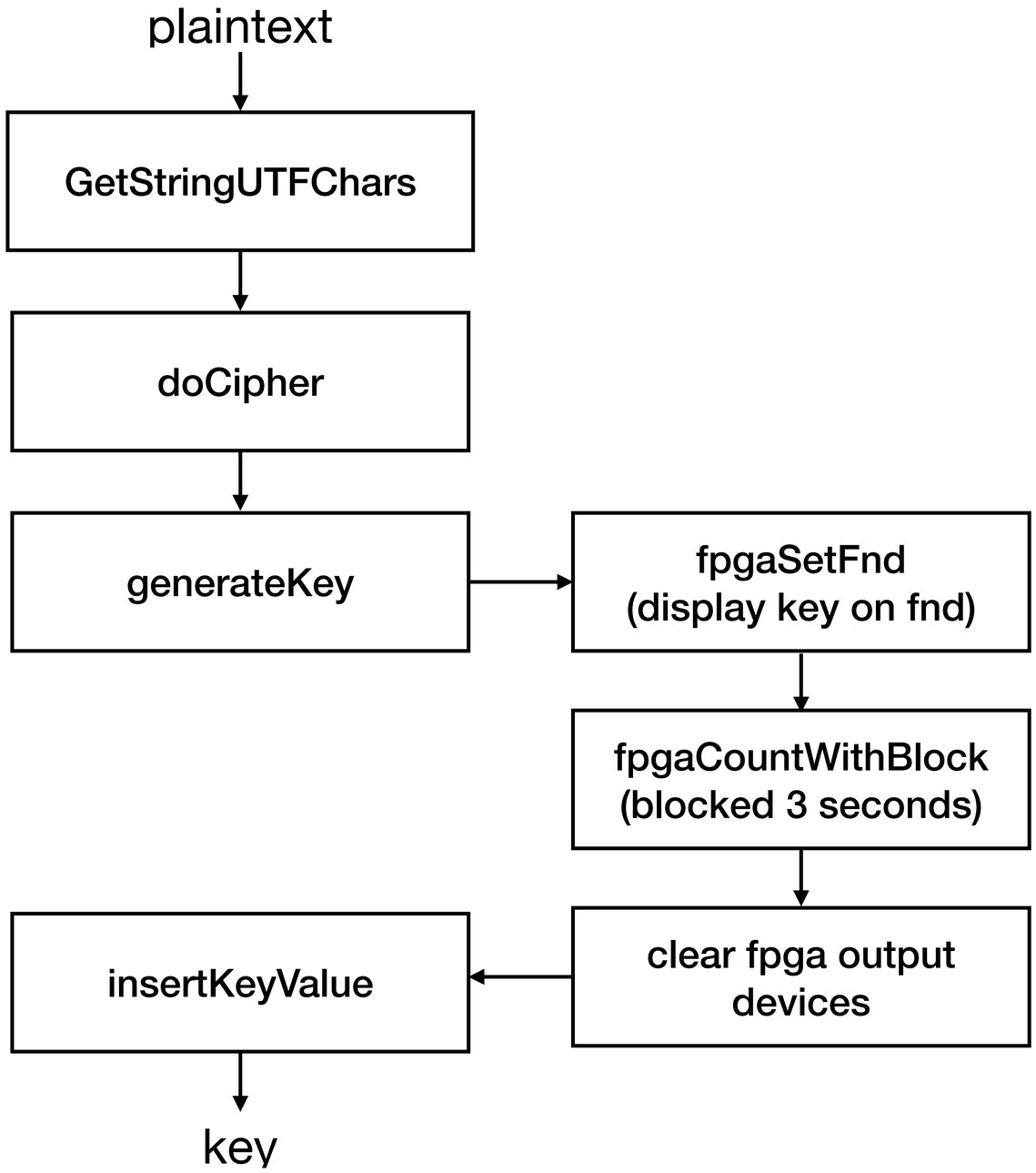
JNI

data structure



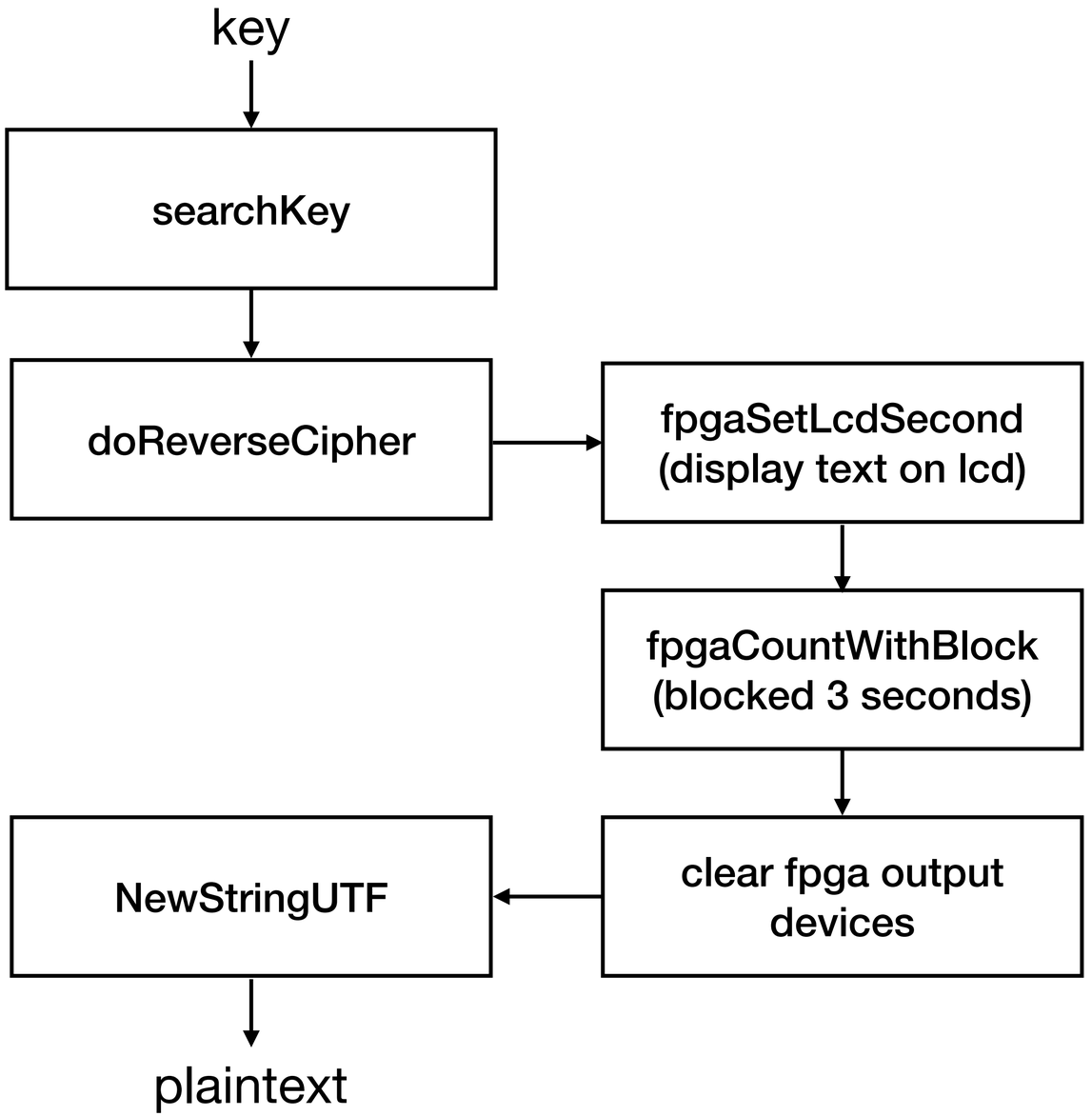
JNI

operation - insert



JNI

operation - search



JNI

fpga function

ioctl command	description
fpgaReadText	fpga device로부터 텍스트를 얻는다.
fpgaReadNumber	fpga device로부터 숫자를 얻는다.
fpgaSetFnd	fnd에 key를 출력한다.
fpgaSetDot	dot에 데이터를 출력한다.
fpgaSetLcdFirst	lcd의 첫번째 줄에 메시지를 출력한다.
fpgaSetLcdSecond	lcd의 두번째 줄에 데이터를 출력한다.
fpgaCountWithBlock	최대 9 초간 block된다.
fpgaClearDevice	fpga output device를 모두 초기화한다.

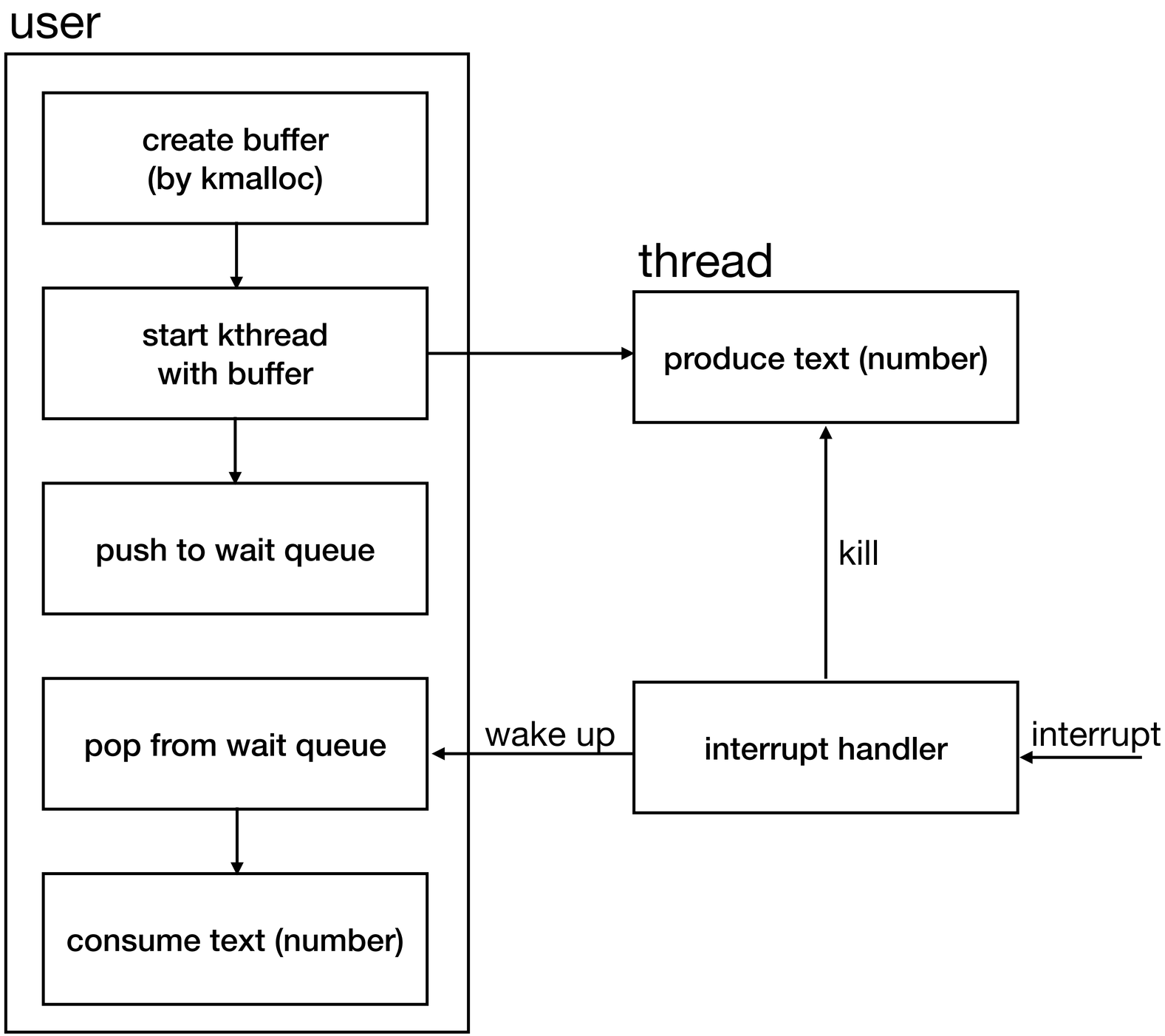
Module

ioctl command

ioctl command	description
GET_TEXT	switch 입력으로부터 텍스트를 추출한다.
GET_NUMBER	switch 입력으로부터 4자리 정수를 추출한다.
SET_FND	fnd에 표시되는 숫자를 수정한다.
SET_DOT	dot에 표시되는 숫자를 수정한다.
SET_LCD_FIRST	text lcd의 첫번째 줄을 수정한다.
SET_LCD_SECOND	text lcd의 두번째 줄을 수정한다.
COUNT_WITH_BLOCK	user process을 최대 9초 간 block 한다.
DEVICE_CLEAR	모든 output device을 초기화한다.

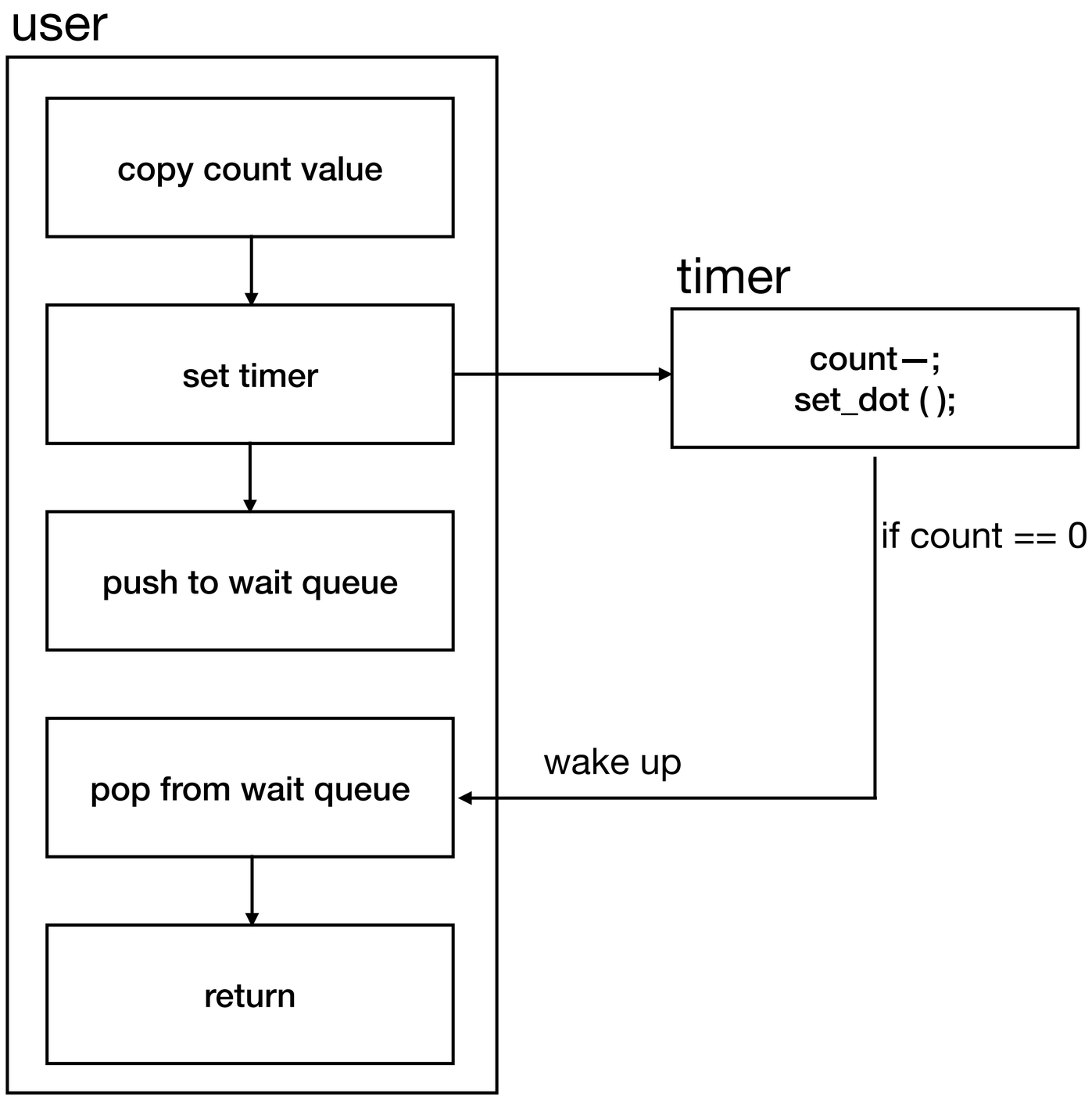
Module

read from switch



Module

count with blocking



감사합니다